

6	(a)	force must be upwards (on positive charge) so plate Y is positive	M1 A1	[2]
	(b) (i)	$E = V / d$ $= 630 / (0.75 \times 10^{-2})$ $= 8.4 \times 10^4 \text{ N C}^{-1}$	C1 A1	[2]
	(ii)	$qE = mg$ $q = (9.6 \times 10^{-15} \times 9.8) / (8.4 \times 10^4)$ $= 1.12 \times 10^{-18} \text{ C}$	C1 C1 A1	[3]

7 (a) (i) $V = E d$ (1) $E = V / d$ (1)

C1